NON-INTERSECTING ARITHMETIC PROGRESSIONS: THE DE LA BRETÈCHE–FORD–VANDEHEY CONJECTURE

CLAUDE OPUS 5.5

Abstract. Let $f(x)$ be the largest number of distinct moduli $q \le x$ for which one can choose residue classes $a_q \pmod q$ that are pairwise disjoint. De la Bretèche, Ford and Vandehey proved in 2013 that $xL(x)^{-1+o(1)} \le f(x) \le xL(x)^{-\sqrt{3}/2+o(1)}$, where $L(x) = \exp\sqrt{\log x \log\log x}$, and conjectured that the lower bound is sharp on this exponential scale. We prove that conjecture: $f(x) = xL(x)^{-1+o(1)}$. This determines the leading constant in $\log(x/f(x))$ in Erdős Problem #202. It also gives $\varepsilon_m = L(m)^{-1+o(1)}$, where ε_m is the supremum of the reciprocal sums of distinct moduli exceeding m that support pairwise disjoint residue classes (Erdős Problem #1190). The proof is the descending-chain argument of de la Bretèche, Ford and Vandehey with one step changed: where they used an Erdős–Lovász-type bound for set-minimal intersecting families, we use a dense-core bound of Alweiss, Lovett, Wu and Zhang, a consequence of the spread lemma from the recent work on the Erdős–Rado sunflower problem: every intersecting family of k-element sets has a non-empty set C lying in at least a $(C_0 \log ek)^{-|C|}$ proportion of its members.

About this version. This paper was written by Claude Opus 5.5 (Anthropic) in October 2026. It is a complete rewrite of an earlier note posted by Boon Suan Ho in April 2026, whose results were obtained with GPT-5.4 Pro. **The original version of this paper is available from `https://boonsuan.github.io/erdos202_old.pdf`.** The theorem and the proof strategy are those of the original. This version adds a proof overview, figures and a fuller history. It identifies the dense-core lemma, presented in the original as a new ingredient derived from the Park–Pham theorem, as Theorem 4.2 of Alweiss, Lovett, Wu and Zhang, a consequence of the spread lemma from the sunflower literature, and it credits de la Bretèche, Ford and Vandehey's own reduction (their Theorem 2) of the problem to a dense-core statement. Two routine estimates are proved more directly.

1. Introduction

1.1. **The problems.** Erdős asked how many residue classes with *distinct* moduli can be pairwise disjoint. In the form recorded on the Erdős problems website [3] the question reads as follows.

Problem (Erdős Problem #202)**.** *Let $n_1 < \cdots < n_r \le N$, with residue classes $a_i \pmod{n_i}$ chosen so that every integer lies in at most one of them. How large can r be in terms of N?*

The problem appears in many of Erdős's problem papers [9, 10, 11, 12, 13, 14, 15] (see also [31, Problem 1.29]). Writing x for N, let $f(x)$ denote the largest possible r. Two residue classes $a \pmod q$ and $b \pmod{q'}$ meet if and only if $a \equiv b \pmod{\gcd(q, q')}$ (Lemma 4.1 below), so $f(x)$ is the largest size of a set $\mathcal{Q} \subseteq [1, x]$ of integers carrying residues $(a_q)_{q\in\mathcal{Q}}$ with

$$a_q \not\equiv a_{q'} \pmod{\gcd(q, q')} \qquad \text{for all distinct } q, q' \in \mathcal{Q}.$$

Date: October 2026.

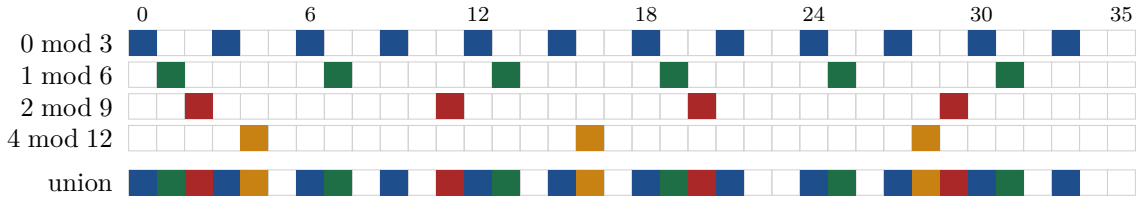


Figure 1. Four pairwise disjoint residue classes, shown over one period $0 \le n < 36$. In the first four rows, no column contains two coloured cells; the last row shows their union, of density $\frac{1}{3}+\frac{1}{6}+\frac{1}{9}+\frac{1}{12} = \frac{25}{36}$. In general, disjoint residue classes differ modulo some prime power dividing both moduli, so their moduli have intersecting prime supports.

We call such a set of moduli *admissible*. Figure 1 shows an admissible set of four moduli up to 12; by the exhaustive computations recorded in OEIS A389975 [25], $f(12) = 4$.

A companion problem asks for the largest possible sum of reciprocals of the moduli, once all moduli exceed m.

Problem (Erdős Problem #1190 [4]; [14, p. 96])**.** *Let* $\varepsilon_m = \sup \sum_{i=1}^{k} 1/n_i$*, the supremum being over all finite sequences* $m < n_1 < \cdots < n_k$ *for which there exist pairwise disjoint residue classes* $a_i \pmod{n_i}$*. Estimate* ε_m*.*

Disjoint classes have total density at most 1, so $\varepsilon_m \le 1$. Erdős [14] attributes $\varepsilon_m < 1$ to Mirsky and Newman, without a reference, and says that he could not decide whether $\varepsilon_m \to 0$.

1.2. **Results.** Throughout, logarithms are natural, $\log_2 x = \log\log x$, and

$$L(\alpha, x) := \exp\bigl(\alpha\sqrt{\log x \, \log_2 x}\bigr), \qquad L(x) := L(1, x).$$

The main result confirms a conjecture of de la Bretèche, Ford and Vandehey [5, Conjecture 1] (henceforth BFV).

Theorem 1.1. *As* $x \to \infty$*,*

$$f(x) = xL(-1 + o(1), x) = x\exp\bigl\{-(1 + o(1))\sqrt{\log x \, \log_2 x}\bigr\}.$$

Only the upper bound is new; the lower bound is BFV's construction [5, Section 2]. Partial summation and truncation give the following asymptotic for the quantity in Problem #1190.

Corollary 1.2. *As* $m \to \infty$*,* $\varepsilon_m = L(-1 + o(1), m) = \exp\bigl\{-(1 + o(1))\sqrt{\log m \, \log_2 m}\bigr\}$*.*

BFV's bounds for f already imply, by the same two arguments (Section 7), that $L(m)^{-1+o(1)} \le \varepsilon_m \le L(m)^{-\sqrt{3}/2+o(1)}$, as recorded on the problem page [4]; in particular $\varepsilon_m \to 0$. Corollary 1.2 identifies the exponent as 1.

1.3. **Background.** Erdős and Stein conjectured that $f(x) = o(x)$, and Erdős and Szemerédi [18] proved this in 1968. Croot [8] brought both bounds to the L-scale and introduced the descending-chain method used below. Table 1 records the bounds; BFV conjectured [5, Conjecture 1] that their lower bound is sharp, which is Theorem 1.1.

1.4. **What is new, and what is not.** The upper-bound argument follows BFV's pruning [5, Section 4.1], descending chain [5, Section 4.2] (which they credit to Croot) and final optimisation [5, Section 4.3], with the core estimate replaced. BFV also showed [5, Theorem 2] that their conjecture would follow from a purely combinatorial "dense-core" statement about intersecting families [5, Conjecture 2]; see Remark 3.4. The new point, due to the original note [20], is

	lower bound for $f(x)$	upper bound for $f(x)$	remark
Erdős–Szemerédi 1968 [18]	$x\exp\{-(\log x)^{1/2+\varepsilon}\}$	$x(\log x)^{-c}$	
Croot 2003 [8]	$xL(x)^{-\sqrt{2}+o(1)}$	$xL(x)^{-1/6+o(1)}$	$xL(x)^{-1/2+o(1)}$ if squarefree
Chen 2005 [6]	—	$xL(x)^{-1/2+o(1)}$	
BFV 2013 [5]	$xL(x)^{-1+o(1)}$	$xL(x)^{-\sqrt{3}/2+o(1)}$	conjecture: lower one sharp
This paper	—	$xL(x)^{-1+o(1)}$	

Table 1. Bounds for $f(x)$ as $x \to \infty$. In the first row $\varepsilon > 0$ is arbitrary and fixed, the lower bound holds up to a constant factor depending on ε, and $c > 0$ is an absolute constant. Croot's remark is his upper bound for admissible sets of squarefree moduli. A dash means that no new bound is recorded.

that a *size-dependent* dense-core bound, whose loss grows only logarithmically with the size of the sets, already suffices for BFV's argument. Such a bound is Theorem 4.2 of Alweiss, Lovett, Wu and Zhang [1], a consequence of their spread lemma in the sharpened form of Rao [27] (see also [28, 2]); we recall it as Corollary 3.3. What has to be checked is that its loss, accumulated over the chain, is negligible. Later developments on the problem forum, including a computer-checked proof, are described in Section 9.

Section 2 explains the argument informally. Section 3 records the spread lemma and the dense-core bound, Section 4 collects the number-theoretic input from BFV, Section 5 runs the descending chain, Section 6 carries out the final optimisation, and Section 7 proves Corollary 1.2. Section 8 contains further remarks on the dense-core bound.

2. Overview of the proof

From disjoint classes to intersecting set systems. Let $\mathcal{Q} \subseteq [1, x]$ be admissible with residues (a_q). Suppose that a subfamily $\mathcal{Q}^*$ of moduli all have the same exact divisor P (that is, $P \mid q$ and $\gcd(P, q/P) = 1$), and that their residues all agree modulo P. For $q \in \mathcal{Q}^*$ let $A(q)$ be the set of primes dividing q/P. If $A(q)$ and $A(q')$ were disjoint, then $\gcd(q, q')$ would divide P, so $a_q \equiv a_{q'} \pmod{\gcd(q, q')}$ and the two classes would meet. Hence $\{A(q) : q \in \mathcal{Q}^*\}$ is an *intersecting* family of sets of primes.

The descending chain. BFV, following Croot, exploit this repeatedly (Fig. 2). After some pruning, all moduli have exactly K distinct prime factors and distinct squarefree kernels. Starting from $P = 1$, at step r one

(i) finds a non-empty set C_r of primes contained in many of the supports $A(q)$, and keeps those moduli;

(ii) fixes the exact prime-power part P_r of q on C_r by weighted pigeonholing, retaining at least a proportion $(6/\pi^2)^{|C_r|}h(P_r)^{-2}$ of the moduli remaining after step (i), where $h(P_r)$ is the product of its prime exponents;

(iii) fixes the residue modulo P_r, keeping the most popular one (this costs a factor P_r);

and replaces P by PP_r. After $R \le K$ steps every prime is fixed and a single modulus $P_1 \cdots P_R \ge x^{1-o(1)}$ survives.

Each step gives an upper bound for the new block P_r, by counting the family $\mathcal{Q}_r$ obtained after fixing the prime powers but before fixing the residue modulo P_r. Its lower bound, obtained by following the losses from the pruned family of size $S' = xL(-\sigma, x)$, has paid only the earlier residue costs $P_1 \cdots P_{r-1}$. Its upper bound is the number of $m \le x/(P_1 \cdots P_r)$ with the right number of prime factors, which is small by a Rankin-type bound, and it contains $P_1 \cdots P_r$. Cancelling the common factor $P_1 \cdots P_{r-1}$ therefore bounds P_r. Multiplying over r and using $\prod P_r \ge x^{1-o(1)}$ gives a lower bound for σ.

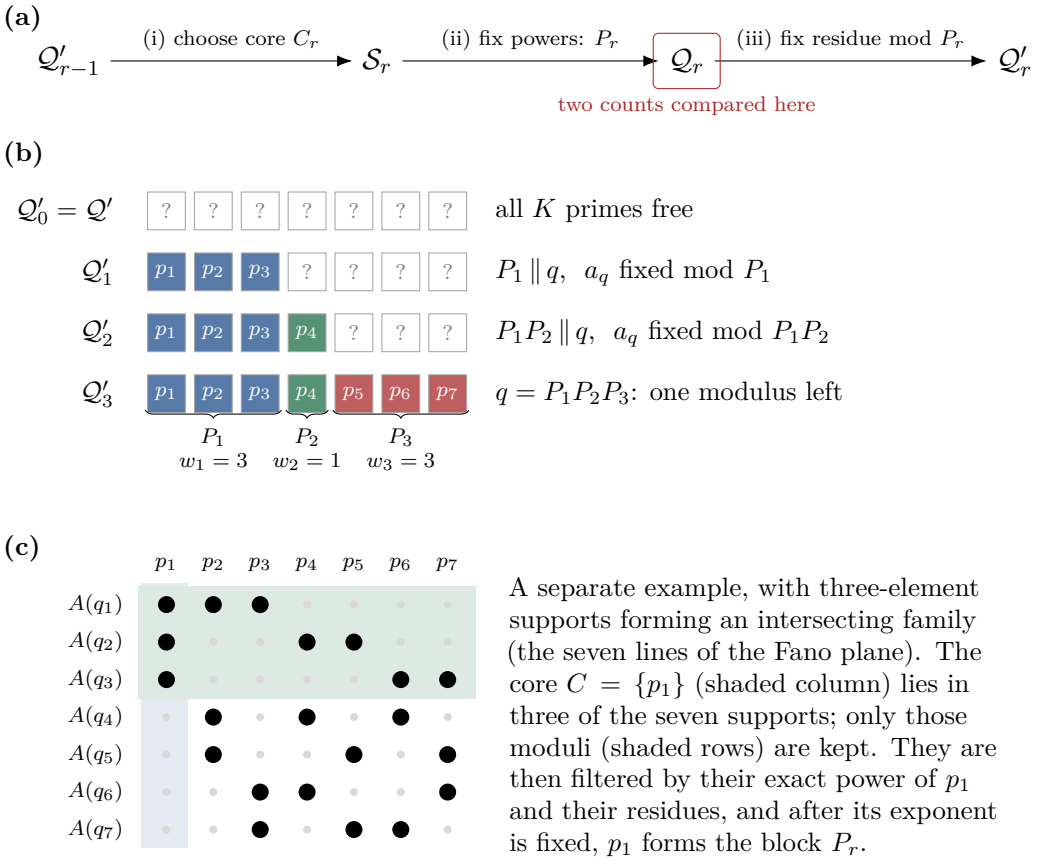


Figure 2. The descending chain of Section 5, schematically. (a) The families of moduli in step r; the lower and upper counts of $\mathcal{Q}_r$ are compared to bound P_r. (b) A surviving modulus at stages $r = 0, 1, 2, 3$ when $K = 7$: coloured cells are the primes already fixed, with their exact exponents, in the blocks P_1, P_2, P_3, and "?" marks primes that still vary over the surviving family. Here $D \,\|\, q$ means that $D \mid q$ and $\gcd(D, q/D) = 1$. At stage r all residues a_q agree modulo $P_1 \cdots P_r$. (c) The core-selection part (i) of a step, in a separate example.

Where the loss is. Write $X = \log x$, $Y = \log_2 x$, and measure the number of prime factors and the number of steps on the scale $M = \sqrt{X/Y}$: $K = dM$ and $R = cM$ (we shall see $0 < c \le d \le 3$). Two of the losses in the chain are intrinsic: the Rankin-type count and the residue step together produce, after taking logarithms and dividing by $\log x$, the quadratic expression $c\sigma - c^2/4$, and the chain argument gives

$$1 \le c\sigma - \tfrac{c^2}{4} + (\text{loss from step (i)}) + o(1).$$

If step (i) cost nothing, then $1 \le c\sigma - c^2/4 \le \sigma^2$ would give $\sigma \ge 1$, which is the conjecture. The question is therefore what fraction of the moduli one can keep in step (i).

BFV shrink the supports to a set-minimal intersecting family, meaning that no member can be replaced by a proper subset while preserving all intersections, and use a bound in the spirit of Erdős and Lovász [16, p. 621]: a set-minimal intersecting family of sets of size at most K has at most wK^{w-1} members of size at most w [5, Lemma 3.4]. This finds a w-set C contained in at least a $4^{-w}K^{-(w-1)}$ proportion of the supports. Since $\log K \le \frac{1}{2}\log_2 x$ for large x, this loss

can be of the same size as the main terms: it adds up to $\frac{c(d-c)}{2}$ to the right-hand side above, and after optimisation leads to the exponent $\sqrt{3}/2$ (Remark 6.1, Fig. 3).

The spread lemma (Theorem 3.2) says that a family of k-sets in which no set C is contained in more than a $\kappa^{-|C|}$ proportion of the members contains two disjoint members, as soon as $\kappa \geq C_0 \log(ek)$. Applied to an intersecting family, it produces a set C in at least a $(C_0 \log eK)^{-|C|}$ proportion of the supports (Corollary 3.3). Together with step (ii), each prime of a core now costs a factor at most $\Lambda := \frac{\pi^2}{6} C_0 \log(eK)$. A core chosen early is paid for again in every later inequality, since the lower bound for $|\mathcal{Q}_r|$ carries all the losses $\Lambda^{|C_1|+\cdots+|C_r|}$ incurred so far. But with $W_r = |C_1| + \cdots + |C_r| \leq K$, the accumulated cost is

$$\log\Big(\prod_{r=1}^{R} \Lambda^{W_r}\Big) = \Big(\sum_{r=1}^{R} W_r\Big) \log \Lambda \leq RK \log \Lambda = O\Big(\frac{X \log Y}{Y}\Big) = o(X),$$

because $RK \leq 9M^2 = 9X/Y$ and $\log \Lambda = O(\log \log K) = O(\log Y)$. With BFV's bound the corresponding cost is about $K^{W_r - r}$ in step r; when $\log K$ is comparable to Y, its accumulation can be comparable to X.

3. Spread families and a dense core

For a finite family $\mathcal{F}$ of finite sets and a set T we write $\mathcal{F}_T := \{F \in \mathcal{F} : T \subseteq F\}$.

Definition 3.1 (Spread families)**.** Let $\kappa \geq 1$. A non-empty finite family $\mathcal{F}$ of finite sets is *κ-spread* if

$$|\mathcal{F}_T| \leq \kappa^{-|T|} |\mathcal{F}| \qquad \text{for every non-empty set } T.$$

A sufficiently spread family contains several pairwise disjoint members; the next theorem quantifies "sufficiently" in terms of the maximum set size.

Theorem 3.2 (Spread lemma; Alweiss–Lovett–Wu–Zhang, Rao, Tao)**.** *There is an absolute constant C_1 with the following property. Let $r \geq 2$ and $k \geq 1$ be integers and let $\mathcal{F}$ be a κ-spread family of non-empty sets, each of size at most k, with $\kappa \geq C_1 r \log(2rk)$. Then $\mathcal{F}$ contains r pairwise disjoint members.*

Corollary 3.3 (Dense core; [1, Theorem 4.2])**.** *There is an absolute constant $C_0 \geq 1$ with the following property. Let $K \geq 1$ and $1 \leq k \leq K$ be integers, and let $\mathcal{A}$ be a non-empty finite intersecting family of distinct k-element sets. Then some non-empty set C satisfies*

$$|\mathcal{A}_C| > \frac{|\mathcal{A}|}{(C_0 \log(eK))^{|C|}}. \tag{1}$$

Proof. Let $C_0 := 4C_1$. If (1) failed for every non-empty C, then $\mathcal{A}$ would be κ-spread with $\kappa = C_0 \log(eK) \geq 4C_1 \log(ek) \geq 2C_1 \log(4k)$ (the last inequality is $4k \leq e^2 k^2$). By Theorem 3.2 with $r = 2$, $\mathcal{A}$ would contain two disjoint members, which are distinct since $k \geq 1$. This contradicts the assumption that $\mathcal{A}$ is intersecting. □

Corollary 3.3 is the contrapositive of Theorem 4.2 of Alweiss, Lovett, Wu and Zhang [1], which states that an intersecting k-uniform family $\mathcal{F}$ with $|\mathcal{F}_T| \leq \kappa^{-|T|}|\mathcal{F}|$ for all T has $\kappa = O(\log k)$, and whose proof uses Rao's sharpened spread estimate; the only adjustments are $\log(ek)$ in place of $\log k$, to cover $k = 1$, and the replacement of k by $K \geq k$. We include the deduction to make the input to the chain explicit. It is the familiar "core or spread" dichotomy behind the sunflower bounds (compare [1, Lemma 2.4] and Lemma 2 of [28], which go back to the argument of Erdős and Rado [17]), which also underlies the spread approximation method of Kupavskii and Zakharov [21]. Up to the value of the constant, the bound in Corollary 3.3 cannot be improved much: see Remark 8.1.

Notes on the literature. Definition 3.1 is the normalisation of Frankston, Kahn, Narayanan and Park [19, (4)] and of Tao [28]; Alweiss, Lovett, Wu and Zhang [1, Definition 1.10], who introduced the term, add the condition $|\mathcal{F}| \geq \kappa^k$ for k-uniform $\mathcal{F}$, which is automatic here (take $T \in \mathcal{F}$). Rao [27] and Bell–Chueluecha–Warnke [2] use a closely related "absolute" normalisation. In exactly the normalisation of Definition 3.1, Theorem 3.2 is Proposition 3 ("locating the petals") of Tao's exposition [28] (stated there for $k \geq 2$, which is no loss, since sets of size at most 1 also have size at most 2). It is the key step in the proofs of the improved sunflower bounds: Alweiss, Lovett, Wu and Zhang [1, Theorem 2.5 and Lemma 1.6] proved it with κ of order $r^2(\log k \log\log k + (\log r)^2)$, Rao [27, Lemma 2] with κ of order $r\log(rk)$, and Bell, Chueluecha and Warnke [2, Lemma 2] with κ of order $r\log k$. In each case the disjoint members are found by colouring the ground set at random ([1, Lemma 1.6]). For $r = 2$, the sharpened forms have order $\log k$. The original estimate of Alweiss, Lovett, Wu and Zhang has an additional factor $\log\log k$, but it too would suffice for the argument below: it would replace $\Lambda \asymp \log K$ in Section 5 by $\Lambda \asymp \log(eK)\log\log(e^eK)$, which still has $\log\Lambda = O(\log\log(eK))$, so that its cumulative contribution in Section 6 remains $o(\log x)$.

Remark 3.4 (Relation to BFV's Conjecture 2). BFV conjectured [5, Conjecture 2] that there are constants $t(1), t(2), \dots$ with $\log t(j) = o(j\log j)$ such that every finite intersecting family $\mathcal{A}$ of non-empty sets has a non-empty C with $|\mathcal{A}_C| \geq |\mathcal{A}|/t(|C|)$, and proved [5, Theorem 2] that this implies $f(x) = xL(-1+o(1), x)$. They remarked that the conclusion holds with $t(j) \ll j^{j+2}$, which is just too weak. Corollary 3.3 supplies $t(j) = (C_0\log eK)^j$ for k-uniform families with $1 \leq k \leq K$. This is not Conjecture 2, because it depends on K. It is nonetheless enough: BFV's proof of their Theorem 2 only uses the bound $\sum_j \log t(w_j) = o(\sqrt{\log x \log_2 x})$ for block sizes with $\sum_j w_j = K \ll \sqrt{\log x/\log_2 x}$, and here $\sum_j \log t(w_j) = K\log(C_0\log eK) = O(K\log\log K)$. The argument below is a self-contained version of this.

4. Input from de la Bretèche, Ford and Vandehey

Notation. From now on x is large and

$$X := \log x, \qquad Y := \log_2 x, \qquad M := \sqrt{X/Y}, \qquad Z := \sqrt{XY},$$

so that $L(\alpha, x) = e^{\alpha Z}$, $Z = MY$ and $X = MZ$. We write $\omega(n)$ for the number of distinct prime factors of n, $\mathrm{rad}(n)$ for its squarefree kernel, $v_p(n)$ for the exponent of p in n, and, following BFV,

$$h(n) := \prod_{p^\nu \,\|\, n} \nu.$$

We write $D \,\|\, n$ ("D is an exact divisor of n") if $D \mid n$ and $\gcd(D, n/D) = 1$; then $h(n) = h(D)\,h(n/D) \geq h(D)$. Every $o(1)$ denotes a function of x alone that tends to 0 as $x \to \infty$; in particular it is uniform in all other parameters.

Lemma 4.1 (Chinese remainder criterion)**.** *The classes $a \pmod q$ and $b \pmod{q'}$ have a common element if and only if $a \equiv b \pmod{\gcd(q, q')}$.*

Proof. A common element n gives $a \equiv n \equiv b \pmod{\gcd(q,q')}$. Conversely, if $\gcd(q,q') \mid a-b$, write $a - b = uq + u'q'$ (Bézout); then $n = a - uq = b + u'q'$ is common to both classes. □

The first estimate is a Rankin-type bound.

Lemma 4.2 ([5, Lemma 3.1])**.** *Fix $A > 0$. Uniformly for $2 \leq y \leq x$ and $0 \leq \alpha \leq A$,*

$$\#\{n \leq y : \omega(n) \geq \alpha M\} \leq yL(-\alpha/2 + o(1), x).$$

Proof (following BFV). Let $z := \sqrt{X}/Y$, so $z \ge 1$ for large x, and put $s := 1 + 1/X$. Since $y^{1/X} \le e$, $\zeta(s) \le s/(s-1) \le 2X$, and $1 + z\sum_{\nu\ge1} p^{-\nu s} \le (1-p^{-s})^{-z}$ for $z \ge 1$,

$$\#\{n \le y : \omega(n) \ge \alpha M\} \le z^{-\alpha M} \sum_{n\le y} z^{\omega(n)} \le z^{-\alpha M} y^s \sum_{n\ge1} \frac{z^{\omega(n)}}{n^s} \le e\, y\, z^{-\alpha M} (2X)^z.$$

As $\log z = \frac12 \log X - \log Y = \frac{Y}{2} - \log Y$ and $\alpha M \cdot \frac{Y}{2} = \frac{\alpha Z}{2}$, the logarithm of the factor after y is $-\alpha Z/2 + O_A(M \log Y + \sqrt{X})$, which is $(-\alpha/2 + o(1))Z$ uniformly, because $M \log Y = Z \log Y / Y$ and $\sqrt{X} = Z/\sqrt{Y}$. □

We use it in the following form.

Lemma 4.3. *There is a function $\eta(x) \ge 0$ tending to 0 such that, for all integers $0 \le W \le K \le 3M$ and all $1 \le y \le x$,*

$$\#\{n \le y : \omega(n) = K - W\} \le y\, L\Big(-\frac{K}{2M} + \eta(x), x\Big)(\log x)^{W/2}.$$

Proof. Let $\eta(x) \ge 0$ be the $o(1)$ of Lemma 4.2 with $A = 3$ (made non-negative). If $W = K$ the left side is $\#\{n \le y : n = 1\} \le 1 \le y$, while the right side is $y\, e^{\eta Z} \ge y$, since $(\log x)^{K/2} = e^{KY/2} = L(K/(2M), x)$. If $W < K$ and $y < 2$ the left side is 0. If $W < K$ and $y \ge 2$, apply Lemma 4.2 with $\alpha = (K-W)/M \in (0, 3]$ to the larger set $\{n \le y : \omega(n) \ge K - W\}$ and note $L(W/(2M), x) = e^{WZ/(2M)} = e^{WY/2} = (\log x)^{W/2}$. □

The second input is BFV's pruning of an extremal family. Fix an admissible $\mathcal{Q} \subseteq [1, x]$ with $|\mathcal{Q}| = f(x) =: S$, and residues $(a_q)_{q\in\mathcal{Q}}$. We retain almost all of this family on the L-scale (P1) while imposing four conditions. Condition (P2) makes the terminal modulus of the chain large; (P3) controls the loss from choosing prime powers in step (ii); (P4) permits a uniform count of the remaining prime factors; and (P5) ensures that counting prime supports is the same as counting moduli.

Proposition 4.4 (Pruning; [5, Section 4.1])**.** *There are a subfamily $\mathcal{Q}' \subseteq \mathcal{Q}$ and an integer K such that, with $S' := |\mathcal{Q}'|$,*

(P1) *$S' \ge S\, L(-o(1), x)$;*
(P2) *$xL(-2, x) \le q \le x$ for every $q \in \mathcal{Q}'$;*
(P3) *$h(q) \le e^{\sqrt{X}}$ for every $q \in \mathcal{Q}'$;*
(P4) *$\omega(q) = K$ for every $q \in \mathcal{Q}'$, and $1 \le K \le 3M$;*
(P5) *the kernels $\mathrm{rad}(q)$, $q \in \mathcal{Q}'$, are distinct.*

Proof sketch. These are conditions (1)–(5) of [5, Section 4.1] (where rad is called ker); we recall why each costs little. By BFV's lower bound, $S \ge xL(-1 - o(1), x)$. Discarding $q < xL(-2, x)$ removes $o(S)$ moduli. By Lemma 4.2 with $\alpha = 3$, at most $xL(-\frac32 + o(1), x) = o(S)$ moduli have $\omega(q) \ge 3M$. By [5, Lemma 3.2], at most $x \exp(-\frac15 \sqrt{X}\, Y) = o(S)$ integers $n \le x$ have $h(n) \ge e^{\sqrt{X}}$ (note $\sqrt{X}\, Y/Z = \sqrt{Y} \to \infty$). Pigeonholing the value of $\omega(q)$ costs a factor at most $3M = L(o(1), x)$. Finally, [5, Lemma 3.3] shows that at most $H^2(\pi^2/6)^K \le H^2 2^K$ integers n with $h(n) \le H$ share a given squarefree kernel with K prime factors; with $H = e^{\sqrt{X}}$ this is $e^{2\sqrt{X}} 2^{3M} = L(o(1), x)$, and keeping one modulus per kernel gives (P5). Since $xL(-2, x) > 1$, (P2) forces $K \ge 1$. □

5. The descending chain

Fix $\mathcal{Q}'$, K and S' as in Proposition 4.4, put $d := K/M \in (0,3]$, let C_0 be the constant of Corollary 3.3, and set

$$\Lambda := \frac{\pi^2}{6}\, C_0 \log(eK).$$

The following is BFV's descending chain [5, Section 4.2], with the dense-core step changed.

Proposition 5.1 (Chain inequality)**.** *There are an integer $1 \le R \le K$ and pairwise coprime integers $P_1, \dots, P_R > 1$ such that, writing $P_{\le r} := P_1 \cdots P_r$ and $W_r := \omega(P_{\le r})$,*

(a) *$W_R = K$ and $P_{\le R} \in \mathcal{Q}'$; in particular $P_{\le R} \ge xL(-2,x)$;*

(b) *for every $1 \le r \le R$,*

$$P_r \le \frac{x}{S'}\, L\Big(-\frac{d}{2} + \eta(x), x\Big)(\log x)^{W_r/2}\, \Lambda^{W_r}\, e^{2\sqrt{X}}, \tag{2}$$

with η as in Lemma 4.3.

Proof. Construction. We construct $P_1, P_2, \dots$ and subfamilies $\mathcal{Q}' = \mathcal{Q}'_0 \supseteq \mathcal{Q}_1 \supseteq \mathcal{Q}'_1 \supseteq \mathcal{Q}_2 \supseteq \cdots$ inductively. Write $P_{<r} := P_{\le r-1}$ (so $P_{<1} = 1$). Suppose that $r \ge 1$ and that a non-empty $\mathcal{Q}'_{r-1} \subseteq \mathcal{Q}'$ has been found such that

$$P_{<r} \,\|\, q \ \text{ for all } q \in \mathcal{Q}'_{r-1}, \qquad \text{and all } a_q\ (q \in \mathcal{Q}'_{r-1}) \text{ agree modulo } P_{<r}. \tag{3}$$

This holds for $r = 1$. If $W_{r-1} = K$ we stop and set $R := r-1$. Otherwise, for $q \in \mathcal{Q}'_{r-1}$ let $A(q)$ be the set of primes dividing $q/P_{<r}$, i.e. the primes of q not dividing $P_{<r}$; then $|A(q)| = K - W_{r-1} =: k$, and $1 \le k \le K$.

The supports are distinct: if $A(q_1) = A(q_2)$ then $\operatorname{rad}(q_1) = \operatorname{rad}(P_{<r}) \prod_{p \in A(q_1)} p = \operatorname{rad}(q_2)$, so $q_1 = q_2$ by (P5).

The supports are intersecting: if $A(q_1) \cap A(q_2) = \emptyset$ for some $q_1 \ne q_2$, then every prime dividing $\gcd(q_1, q_2)$ divides $P_{<r}$, and for such p we have $v_p(q_1) = v_p(P_{<r}) = v_p(q_2)$ by (3); hence $\gcd(q_1, q_2) \mid P_{<r}$. As $a_{q_1} \equiv a_{q_2} \pmod{P_{<r}}$, Lemma 4.1 shows that the two classes meet, contradicting admissibility.

Step (i): dense core. By Corollary 3.3 applied to $\{A(q) : q \in \mathcal{Q}'_{r-1}\}$, there is a non-empty set C_r of primes, of size $w_r := |C_r|$, such that

$$\mathcal{S}_r := \{q \in \mathcal{Q}'_{r-1} : C_r \subseteq A(q)\} \quad \text{satisfies} \quad |\mathcal{S}_r| > \frac{|\mathcal{Q}'_{r-1}|}{(C_0 \log eK)^{w_r}}.$$

Step (ii): exact powers. For $q \in \mathcal{S}_r$ let $B(q) := \prod_{p \in C_r} p^{v_p(q)}$, and for each value u of B let $N_u := \#\{q \in \mathcal{S}_r : B(q) = u\}$. Every value has the form $u = \prod_{p \in C_r} p^{\nu_p}$ with all $\nu_p \ge 1$, and $h(u) = \prod \nu_p$, so

$$\sum_u h(u)^{-2} \le \prod_{p \in C_r} \sum_{\nu \ge 1} \nu^{-2} = \Big(\frac{\pi^2}{6}\Big)^{w_r}.$$

Since $\sum_u N_u = |\mathcal{S}_r|$, some value $u =: P_r$ has $N_{P_r} \ge |\mathcal{S}_r|\, h(P_r)^{-2} (\pi^2/6)^{-w_r}$ (otherwise summing over u gives $|\mathcal{S}_r| < |\mathcal{S}_r|$). Put $\mathcal{Q}_r := \{q \in \mathcal{S}_r : B(q) = P_r\}$. Then

$$|\mathcal{Q}_r| > \frac{|\mathcal{Q}'_{r-1}|}{\Lambda^{w_r}\, h(P_r)^2}. \tag{4}$$

Step (iii): residues. Let $\mathcal{Q}'_r$ be the set of $q \in \mathcal{Q}_r$ whose residue a_q mod P_r is the most frequent one; then

$$|\mathcal{Q}'_r| \ge |\mathcal{Q}_r|/P_r. \tag{5}$$

The invariant is preserved. The primes of P_r lie in C_r, hence do not divide $P_{<r}$, so $P_r > 1$ is coprime to $P_{<r}$. For $q \in \mathcal{Q}'_r$ we have $P_{<r} \,\|\, q$ and $v_p(q) = v_p(P_r)$ for $p \in C_r$, so $P_{\le r} = P_{<r}P_r \,\|\, q$. The residues a_q, $q \in \mathcal{Q}'_r$, agree modulo $P_{<r}$ and modulo P_r, hence modulo $P_{\le r}$. Moreover $\mathcal{Q}'_r \neq \emptyset$ and $W_r = W_{r-1} + w_r > W_{r-1}$, so the process stops after $R \le K$ steps, and $R \ge 1$ because $W_0 = 0 < K$.

Part (a). When the process stops, every $q \in \mathcal{Q}'_R$ satisfies $P_{\le R} \,\|\, q$ and $\omega(q) = K = \omega(P_{\le R})$, so $q/P_{\le R}$ has no prime factor and $q = P_{\le R}$. Thus $\mathcal{Q}'_R = \{P_{\le R}\}$ and $P_{\le R} \in \mathcal{Q}'$, and (P2) gives $P_{\le R} \ge xL(-2,x)$.

Part (b). Fix $1 \le r \le R$. Iterating (4) and (5), and using $h(P_{\le r}) = \prod_{j\le r} h(P_j)$ (the P_j are pairwise coprime) and $W_r = \sum_{j\le r} w_j$,

$$|\mathcal{Q}_r| > S' \prod_{j<r} \frac{1}{P_j} \prod_{j\le r} \frac{1}{\Lambda^{w_j} h(P_j)^2} = \frac{S'}{P_{<r}\,\Lambda^{W_r}\, h(P_{\le r})^2}. \tag{6}$$

On the other hand, each $q \in \mathcal{Q}_r$ is $q = P_{\le r}m$ with $m \le x/P_{\le r}$ and $\omega(m) = K - W_r$, and distinct q give distinct m. Since $1 \le m \le x/P_{\le r} \le x$, Lemma 4.3 gives

$$|\mathcal{Q}_r| \le \frac{x}{P_{\le r}}\, L\Big(-\frac{d}{2} + \eta(x), x\Big)(\log x)^{W_r/2}. \tag{7}$$

Since $P_{\le r} \,\|\, q$ for $q \in \mathcal{Q}_r$, (P3) gives $h(P_{\le r}) \le h(q) \le e^{\sqrt{X}}$. Combining (6) and (7) and cancelling $P_{<r}$ gives (2). □

Remark 5.2. In BFV's version [5, (4.1)] the factor Λ^{w_r} in (4) is $7^{w_r}K^{w_r-1}$. Every other step is the same.

6. Proof of Theorem 1.1

Proof of Theorem 1.1. The lower bound is [5, Section 2]. For the upper bound, take $\mathcal{Q}'$, S', K from Proposition 4.4 and R, $P_1, \dots, P_R$ from Proposition 5.1. Define $\sigma = \sigma(x)$ by $S' = xL(-\sigma, x)$; then $\sigma \ge 0$ because $S' \le x$. Put $c := R/M$ and $d := K/M$, so $0 < c \le d \le 3$.

Multiply (2) over $1 \le r \le R$ and use (a):

$$xL(-2,x) \le P_{\le R} \le \Big(\frac{x}{S'} L\Big(-\frac{d}{2} + \eta, x\Big)\Big)^R (\log x)^{T/2}\, \Lambda^T\, e^{2R\sqrt{X}}, \qquad T := \sum_{r=1}^{R} W_r.$$

Since each later block contains at least one prime, $W_r \le K - (R - r)$, so

$$T \le \sum_{r=1}^{R} (K - R + r) = RK - \frac{R^2}{2} + \frac{R}{2}.$$

Take logarithms, recalling $\log(x/S') = \sigma Z$, $RZ = cX$, $RKY = cdX$ and $R^2Y = c^2X$:

$$\begin{aligned} X - 2Z &\le c\sigma X - \frac{cd}{2}X + c\eta X + \frac{T}{2}Y + T\log\Lambda + 2R\sqrt{X} \\ &\le c\sigma X - \frac{cd}{2}X + c\eta X + \frac{cd}{2}X - \frac{c^2}{4}X + \frac{RY}{4} + T\log\Lambda + 2R\sqrt{X}. \end{aligned}$$

The terms $\pm\frac{cd}{2}X$ cancel. Since $K \le 3M$ and $\Lambda > 1$, we have $0 < \log\Lambda \le \log\Lambda_{\max}$, where $\Lambda_{\max} = \Lambda_{\max}(x) := \frac{\pi^2}{6} C_0 \log(3eM)$. Dividing by X and using $c \le 3$, $R \le 3M$ and $T \le RK \le 9M^2$, we obtain

$$1 - E(x) \le c\sigma - \frac{c^2}{4}, \tag{8}$$

where

$$E(x) := \frac{2Z + 3\eta X + \frac{3}{4}MY + 9M^2 \log \Lambda_{\max} + 6M\sqrt{X}}{X}.$$

Each term of $E(x)$ tends to 0, as $Z/X = MY/X = 1/M$, $\eta \to 0$, $M\sqrt{X}/X = 1/\sqrt{Y}$, and $M^2 \log \Lambda_{\max}/X = \log \Lambda_{\max}/Y$ with $\log \Lambda_{\max} = \log\log(3eM) + O(1) = O(\log Y)$. Now

$$c\sigma - \frac{c^2}{4} = \sigma^2 - \left(\sigma - \frac{c}{2}\right)^2 \le \sigma^2,$$

so (8) gives $\sigma \ge \sqrt{1 - E(x)} = 1 - o(1)$, i.e. $S' \le xL(-1 + o(1), x)$. Finally (P1) gives $f(x) = S \le S'L(o(1), x) \le xL(-1 + o(1), x)$. □

Remark 6.1 (Where $\sqrt{3}/2$ came from). With BFV's core step, (4) has $7^{w_r}K^{w_r-1}$ in place of Λ^{w_r}, so the right side of (2) has $7^{W_r}K^{W_r-r}$ in place of Λ^{W_r}, and the product over r acquires the factor $K^{\sum_r (W_r - r)}$ (the factors 7^{W_r} are as harmless as Λ^{W_r}). Here $\sum_r (W_r - r) \le R(K - R)$ by $W_r \le K - (R - r)$, and $\log K \le Y/2$ for large x (as $K \le 3M$; so BFV). So the additional logarithmic contribution is at most $R(K - R)Y/2$, which after division by X is $\frac{c(d-c)}{2}$, and (8) becomes

$$1 - o(1) \le c\sigma + \frac{cd}{2} - \frac{3c^2}{4}.$$

First, c is bounded away from 0: since $\sigma \le 1 + o(1)$ (by BFV's lower bound and (P1)) and $d \le 3$, the display gives $1 - o(1) \le c(\frac{5}{2} + o(1))$, so $c \ge \frac{2}{5} - o(1)$. Dividing by c therefore gives $\sigma \ge \frac{1}{c} + \frac{3c}{4} - \frac{d}{2} - o(1)$, which is the bound $S' \le xL(-\frac{1}{c} - \frac{3c}{4} + \frac{d}{2} + o(1), x)$ of [5, Section 4.3]. Second, d no longer cancels, so BFV combine this with $\sigma \ge \frac{d}{2} - o(1)$, which follows from Lemma 4.2 as every $q \in \mathcal{Q}'$ has $\omega(q) = dM$. The worst case of $\max\{\frac{1}{c} + \frac{3c}{4} - \frac{d}{2}, \frac{d}{2}\}$ over the admissible range $c \le d \le 3$ is as follows. For $\frac{2}{5} \le c \le 2$ it occurs at $d = \frac{1}{c} + \frac{3c}{4}$, which lies in $[c, 3]$ for these c, and gives $\frac{1}{2}(\frac{1}{c} + \frac{3c}{4}) \ge \frac{\sqrt{3}}{2}$, with equality at $c = 2/\sqrt{3}$. For $2 \le c \le 3$ it occurs at $d = c$ and gives $\frac{c}{2} \ge 1$. Hence BFV's argument yields $\sigma \ge \frac{\sqrt{3}}{2} - o(1)$ and no more. In our argument d cancels, and the second bound is not needed (Fig. 3).

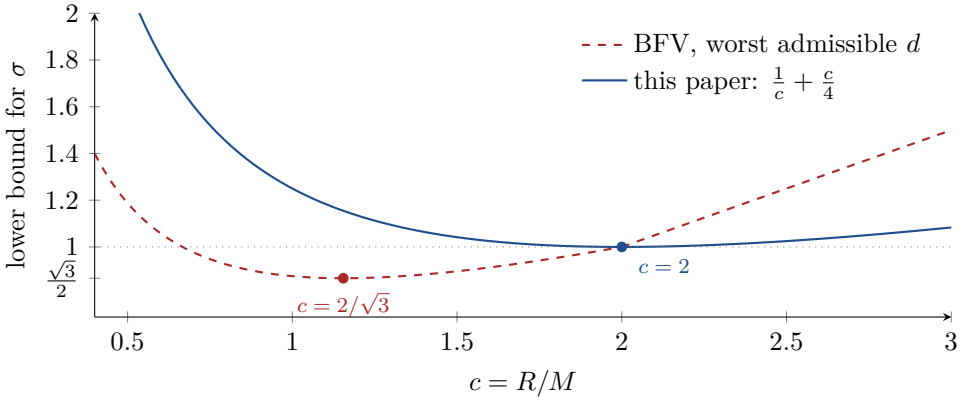


Figure 3. The final optimisation, error terms suppressed ($M = \sqrt{\log x / \log_2 x}$, $|\mathcal{Q}'| = xL(-\sigma, x)$, $R = cM$). As c cannot be chosen, the guaranteed bound is the minimum of a curve. Solid: $\frac{1}{c} + \frac{c}{4}$ from (8), minimum 1 at $c = 2$. Dashed: BFV's bound $\frac{1}{2}(\frac{1}{c} + \frac{3c}{4})$ for $c \le 2$, $\frac{c}{2}$ for $c \ge 2$ (Remark 6.1), minimum $\sqrt{3}/2$ at $c = 2/\sqrt{3}$. For $c > 2$ it lies above the solid curve only because it includes the bound $\sigma \ge \frac{d}{2} \ge \frac{c}{2}$, which is available here too.

Remark 6.2 (Consistency with the construction). The argument can be close to sharp only when $c\sigma - c^2/4 \approx \sigma^2$, i.e. $c \approx 2\sigma$; for $\sigma \approx 1$ the chain must then have about $2M$ steps. BFV's lower-bound construction [5, Section 2] has exactly this shape. There each modulus is a product $p_0 p_1 \cdots p_r$ of $r+1$ primes with $r \sim 2M$: the prime p_0 is common to all moduli, and the residue of a_q modulo p_{k-1} determines p_k. Run on this family, the chain can fix one prime per step, starting with p_0, giving $R = K = r+1 \sim 2M$.

7. Proof of Corollary 1.2

We need one elementary integral estimate.

Lemma 7.1. *For every fixed $\beta > 0$, as $m \to \infty$,*

$$\int_m^\infty \frac{dt}{t\,L(\beta,t)} \le L(-\beta + o(1), m).$$

Proof. Substitute $u = \log t$ and let $g(u) = \sqrt{u \log u}$, which is increasing for $u \ge 1$; with $u_0 = \log m$ the integral is $\int_{u_0}^\infty e^{-\beta g(u)}\,du$. Split $[u_0, \infty)$ into the intervals $[2^j u_0, 2^{j+1} u_0]$, $j \ge 0$. On the j-th interval $g(u) \ge g(2^j u_0) \ge 2^{j/2} g(u_0) \ge (1 + (\sqrt{2}-1)j)\, g(u_0)$, the last step by convexity of $j \mapsto 2^{j/2}$. Hence

$$\int_{u_0}^\infty e^{-\beta g(u)}\,du \le u_0 e^{-\beta g(u_0)} \sum_{j \ge 0} \bigl(2e^{-\beta(\sqrt{2}-1) g(u_0)}\bigr)^j \le 2u_0 e^{-\beta g(u_0)}$$

once $2e^{-\beta(\sqrt{2}-1)g(u_0)} \le \frac{1}{2}$. Since $\log(2u_0) = o(g(u_0))$ and $g(u_0) = \sqrt{\log m \log_2 m}$, the right side is $L(-\beta + o(1), m)$. □

Proof of Corollary 1.2. Upper bound. Fix $\delta \in (0,1)$. By Theorem 1.1 there is t_0 with $f(t) \le tL(-1+\delta, t)$ for $t \ge t_0$. Let $m \ge t_0$ and let $m < n_1 < \cdots < n_k$ carry pairwise disjoint classes. Put $A(t) := \#\{i : n_i \le t\}$. The moduli $n_i \le t$ form an admissible subset of $[1,t]$, so $A(t) \le f(t)$, and $A(t) = 0$ for $t < n_1$. By partial summation,

$$\sum_{i=1}^k \frac{1}{n_i} = \sum_{i=1}^k \int_{n_i}^\infty \frac{dt}{t^2} = \int_m^\infty \frac{A(t)}{t^2}\,dt \le \int_m^\infty \frac{dt}{t\,L(1-\delta,t)} \le L(-1+\delta+o(1), m)$$

by Lemma 7.1. Hence $\varepsilon_m \le L(-1+\delta+o(1), m)$ for every $\delta > 0$, i.e. $\varepsilon_m \le L(-1+o(1), m)$.

Lower bound. Let $x := mL(2,m)$. Then $\log x = (1+o(1))\log m$ and $\log_2 x = \log_2 m + o(1)$, so $L(\alpha, x) = L(\alpha + o(1), m)$ for each fixed α. BFV's construction gives an admissible family $\mathcal{B}_x \subseteq [1,x]$ with $|\mathcal{B}_x| = xL(-1+o(1), x) = mL(1+o(1), m)$. At most m of its members are $\le m$, and $m = o(|\mathcal{B}_x|)$; the remaining $|\mathcal{B}_x|(1-o(1))$ moduli lie in $(m, x]$, so

$$\varepsilon_m \ge \frac{|\mathcal{B}_x|(1-o(1))}{x} = L(-1+o(1), x) = L(-1+o(1), m).$$ □

This direction uses only BFV's construction; with their upper bound for f in place of Theorem 1.1, the upper-bound argument gives $\varepsilon_m \le L(-\sqrt{3}/2 + o(1), m)$.

8. Further remarks on the dense-core bound

Remark 8.1 (A constant exponential bound is impossible). Corollary 3.3 cannot hold with $(C_0 \log eK)^{|C|}$ replaced by $A^{|C|}$ for an absolute constant A. Lovett, Solomon and Zhang [23, Claim 6.8] constructed intersecting k-uniform families that are κ-spread with $\kappa \gg \log k / \log\log k$, using one whole block plus one point from each other block (Alweiss, Lovett, Wu and Zhang note after [1, Theorem 4.2] that this makes their bound nearly sharp). The following specialisation of their construction has explicit counts. For an integer $s \ge 2$ take $t = s^s$ disjoint blocks

$B_1, \dots, B_t$ of size s, and let $\mathcal{A}$ consist of the sets $B_i \cup \{x_l : l \neq i\}$ with $x_l \in B_l$. These are k-sets with $k = s^s + s - 1$, so that $s/2 \sim \frac{1}{2}\log k/\log\log k$, and two members meet, because each contains a point of the other's whole block. We claim that $\mathcal{A}$ is $(s/2)$-spread. If C meets m blocks, each in one point, then $|\mathcal{A}|/|\mathcal{A}_C| = ts^m/(ms + t - m) \geq (s/2)^m$; if C meets one block in $c \geq 2$ points and $m - 1$ others in one point each, then $|\mathcal{A}|/|\mathcal{A}_C| = ts^{m-1} \geq (s/2)^{c+m-1}$ because $t = s^s \geq (s/2)^c$; otherwise $\mathcal{A}_C = \emptyset$. Small instances of this block construction are checked by brute force, and the counts above have been verified exactly by computer for $2 \leq s \leq 5$.

Remark 8.2 (Other routes). Theorem 3.2 with $r = 2$ also follows from the theorem of Frankston, Kahn, Narayanan and Park [19], and from the Park–Pham theorem [26] (the Kahn–Kalai conjecture); the original note [20] took the latter route. Recall the terminology. For a family $\mathcal{F}$ of subsets of a finite ground set V, its *up-closure* is $\langle\mathcal{F}\rangle := \{S \subseteq V : S \supseteq F \text{ for some } F \in \mathcal{F}\}$. A *cover* of an up-closed family $\mathcal{U}$ is a family $\mathcal{G}$ of sets such that every member of $\mathcal{U}$ contains some $G \in \mathcal{G}$. The *threshold* $p_c(\mathcal{U})$ is the p at which a p-random subset of V lies in $\mathcal{U}$ with probability $\frac{1}{2}$, and the *expectation threshold* $q(\mathcal{U})$ is the largest p for which some cover $\mathcal{G}$ of $\mathcal{U}$ has $\sum_{G\in\mathcal{G}} p^{|G|} \leq \frac{1}{2}$. Park and Pham proved that $p_c(\mathcal{U}) \leq K_{\mathrm{PP}}\, q(\mathcal{U}) \log \ell(\mathcal{U})$ for an absolute constant K_{PP}, where $\ell(\mathcal{U}) \geq 2$ bounds the sizes of the minimal members of $\mathcal{U}$. In outline: let $\mathcal{F}$ be κ-spread with sets of size at most k. Any cover $\mathcal{G}$ of $\langle\mathcal{F}\rangle$ satisfies $|\mathcal{F}| \leq \sum_{G\in\mathcal{G}} |\mathcal{F}_G| \leq |\mathcal{F}| \sum_G \kappa^{-|G|}$, so $\sum_G p^{|G|} \geq 1$ for $p \geq \kappa^{-1}$, and $q(\langle\mathcal{F}\rangle) < \kappa^{-1}$. The minimal members of $\langle\mathcal{F}\rangle$ belong to $\mathcal{F}$, so the Park–Pham theorem gives $p_c(\langle\mathcal{F}\rangle) \leq K_{\mathrm{PP}}\kappa^{-1}\log\max(k, 2)$. If this is at most $1/(2r)$, colour the ground set randomly with $2r$ colours: each colour class is a $\frac{1}{2r}$-random set, so it contains a member of $\mathcal{F}$ with probability at least $\frac{1}{2}$, and by linearity of expectation some colouring has r such classes, giving r disjoint members. (The use of $2r$ colours and linearity of expectation, rather than r colours and a union bound, is the device of [2].)

9. Subsequent work

The original note [20] was announced on the forum thread for Problem #202 [29] on 23 April 2026. Thomas Bloom replied that he would update the problem page once the proof had been formalised or checked by a human expert. On 14 May 2026 the forum user *paws* posted a Lean 4/Mathlib formalisation of Theorem 1.1 and Corollary 1.2, in the respective Problem 202 and Problem 1190 proof files [22], depending only on Lean's core axioms; Nat Sothanaphan confirmed it, noting that the Park–Pham input used there was itself formalised rather than assumed. Both problem pages now record the result [3, 4].

On the thread for Problem #1190 [30], MalekZ posted on 28 April 2026 a short note [24] deriving $\varepsilon_m = L(m)^{-1+o(1)}$ from the asymptotic for f by Abel summation and truncation of an extremal family, as in Section 7, and on 30 April 2026 Przemek Chojecki posted a separate write-up produced with GPT-5.5 Pro [7], which proves Theorem 1.1 by essentially the same route (BFV's chain together with the spread lemma), as Sothanaphan observed. Exact values of $f(n)$ for $n \leq 100$ are in OEIS A389975 [25].

Acknowledgements and provenance

The results were found by GPT-5.4 Pro, prompted by Boon Suan Ho, and first written up in [20]; the dense-core bound used there is the known result [1, Theorem 4.2], and the new contribution of the note is its use in BFV's chain. This paper uses comments by Thomas Bloom, paws and Nat Sothanaphan on the forum thread https://www.erdosproblems.com/forum/thread/202, and by MalekZ, Przemek Chojecki and Nat Sothanaphan on https://www.erdosproblems.com/forum/thread/1190. The data for Fig. 1 and the checks in Remark 8.1 were computed by the present author. This text was written by Claude Opus 5.5.

References

[1] R. Alweiss, S. Lovett, K. Wu and J. Zhang, *Improved bounds for the sunflower lemma*, Ann. of Math. (2) **194** (2021), no. 3, 795–815; arXiv:1908.08483.

[2] T. Bell, S. Chueluecha and L. Warnke, *Note on sunflowers*, Discrete Math. **344** (2021), no. 7, Paper No. 112367.

[3] T. F. Bloom, *Erdős Problem #202*, https://www.erdosproblems.com/202, accessed 28 October 2026.

[4] T. F. Bloom, *Erdős Problem #1190*, https://www.erdosproblems.com/1190, accessed 28 October 2026.

[5] R. de la Bretèche, K. Ford and J. Vandehey, *On non-intersecting arithmetic progressions*, Acta Arith. **157** (2013), no. 4, 381–392.

[6] Y.-G. Chen, *On disjoint arithmetic progressions*, Acta Arith. **118** (2005), no. 2, 143–148.

[7] P. Chojecki (with GPT-5.5 Pro), *An unconditional spread-theoretic solution of Erdős Problem #1190*, draft note, 30 April 2026, https://www.ulam.ai/research/erdos1190.pdf.

[8] E. S. Croot III, *On non-intersecting arithmetic progressions*, Acta Arith. **110** (2003), no. 3, 233–238.

[9] P. Erdős, *Some unsolved problems*, Magyar Tud. Akad. Mat. Kutató Int. Közl. **6** (1961), 221–254.

[10] P. Erdős, *Extremal problems in number theory*, in: Proc. Sympos. Pure Math., Vol. VIII, Amer. Math. Soc., Providence, RI, 1965, pp. 181–189.

[11] P. Erdős, *Some recent advances and current problems in number theory*, in: Lectures on Modern Mathematics, Vol. III, Wiley, New York, 1965, pp. 196–244.

[12] P. Erdős, *Problems and results on combinatorial number theory*, in: A Survey of Combinatorial Theory (Fort Collins, 1971), North-Holland, Amsterdam, 1973, pp. 117–138.

[13] P. Erdős, *Problems and results on combinatorial number theory. III*, in: Number Theory Day (New York, 1976), Lecture Notes in Math. 626, Springer, Berlin, 1977, pp. 43–72.

[14] P. Erdős, *A survey of problems in combinatorial number theory*, Ann. Discrete Math. **6** (1980), 89–115.

[15] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, Monographies de L'Enseignement Mathématique 28, Université de Genève, Geneva, 1980.

[16] P. Erdős and L. Lovász, *Problems and results on* 3*-chromatic hypergraphs and some related questions*, in: Infinite and Finite Sets (Keszthely, 1973), Vol. II, Colloq. Math. Soc. János Bolyai 10, North-Holland, Amsterdam, 1975, pp. 609–627.

[17] P. Erdős and R. Rado, *Intersection theorems for systems of sets*, J. London Math. Soc. **35** (1960), 85–90.

[18] P. Erdős and E. Szemerédi, *On a problem of P. Erdős and S. Stein*, Acta Arith. **15** (1968), 85–90.

[19] K. Frankston, J. Kahn, B. Narayanan and J. Park, *Thresholds versus fractional expectation-thresholds*, Ann. of Math. (2) **194** (2021), no. 2, 475–495.

[20] B. S. Ho, *Non-intersecting arithmetic progressions via spread cores*, note, April 2026, https://boonsuan.github.io/erdos202_old.pdf.

[21] A. Kupavskii and D. Zakharov, *Spread approximations for forbidden intersections problems*, Adv. Math. **445** (2024), Paper No. 109653.

[22] paws, *Lean 4 formalisation of Erdős Problems #202 and #1190*, posted 14 May 2026; files `P202/Proof.lean` and `P1190/Proof.lean` at commit `5e507ad`, https://github.com/Shashi456/erdos-formalizations/tree/5e507adea9545aac44880ea5787caf0b0668bf13/Erdos.

[23] S. Lovett, N. Solomon and J. Zhang, *From DNF compression to sunflower theorems via regularity*, in: 34th Computational Complexity Conference (CCC 2019), LIPIcs, Schloss Dagstuhl, 2019, pp. 5:1–5:14; arXiv:1903.00580.

[24] MalekZ, note deriving Problem #1190 from Problem #202, posted 28 April 2026 on [30], https://drive.google.com/file/d/1QcMV27Haw0_jvH17X1H3yofy7D6j1LVQ/view.

[25] OEIS Foundation, *Sequence A389975: maximum cardinality of a set of disjoint congruence classes with distinct moduli, each at most n*, https://oeis.org/A389975.

[26] J. Park and H. T. Pham, *A proof of the Kahn–Kalai conjecture*, J. Amer. Math. Soc. **37** (2024), no. 1, 235–243.

[27] A. Rao, *Coding for sunflowers*, Discrete Anal. 2020, Paper No. 2, 8 pp.
[28] T. Tao, *The sunflower lemma via Shannon entropy*, blog post, 20 July 2020, https://terrytao.wordpress.com/2020/07/20/the-sunflower-lemma-via-shannon-entropy/.
[29] Erdős Problems forum, *Discussion thread for Problem #202*, https://www.erdosproblems.com/forum/thread/202, accessed 28 October 2026.
[30] Erdős Problems forum, *Discussion thread for Problem #1190*, https://www.erdosproblems.com/forum/thread/1190, accessed 28 October 2026.
[31] Various authors, *Some of Paul's favorite problems*, booklet for the conference "Paul Erdős and his mathematics", Budapest, July 1999.

Anthropic